

Udbhav Ram

Medical Physics · Clinical AI · Software

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PUBLIC CV · UPDATED OCTOBER 2026

Education

Ph.D. student, Medical Physics | 2026-Present

University of Wisconsin-Madison, Madison, WI

- Research interests include artificial intelligence for radiation oncology, image segmentation, vision-language models, treatment planning and image-guided radiotherapy.

B.Sc. Honours Medical Physics (with Co-op) | 2021-2026

McMaster University, Hamilton, ON

- Coursework included clinical applications of physics in medicine, radiation biology, and radiation and radioisotope methodology.

Research Experience

Research Assistant, Medical Physics / Radiology | June 2026-Present

University of Wisconsin-Madison, Madison, WI

- Graduate research with mentor Dr. Ran Zhang, with a focus on vision-language models and AI applications in medical imaging and radiation oncology.

International Visiting Scholar and Research Collaborator

Department of Radiation Oncology, University of Alabama at Birmingham, Birmingham, AL

Research collaboration began in 2021; visiting-scholar appointments in 2024 and 2025.

- Evaluated Ethos 2.0 high-fidelity mode for multi-metastasis, single-isocenter stereotactic radiotherapy using semi-automated comparative analysis.
- Investigated inter-observer contouring variability in adaptive accelerated partial breast irradiation and LLM-assisted TG-263 target-name quality assurance.
- Evaluated deep-learning frameworks for abdominal organ auto-segmentation and investigated machine-learning approaches to radiosensitivity and fractionated dose delivery.
- Completed 400 hours of clinical shadowing in radiation oncology and medical physics.

Undergraduate Thesis Student | Sept 2024-May 2025

Juravinski Cancer Centre, Hamilton, ON

- Conducted a JCC-UAB commissioning study comparing 6X and 10X flattening filter-free photon beams on a TrueBeam iTX linac for SBRT using the Eclipse treatment planning system.

Researcher, Department of Medical Physics | Aug 2022-Jan 2023

University of Western Ontario, London, ON

- Built deep-learning models for real-time optical-scan analysis of cerebral hemoglobin oxygenation.

Clinical Research Fellow, Department of Nephrology | Sept 2021-Aug 2022

St. Joseph's Hospital, Hamilton, ON

- Designed deep-learning and computer-vision tools for augmented-reality support in 5/6 nephrectomy mouse models.

Research Experience (continued)

High School Research Fellow, Physics and Astronomy | Sept 2019-Aug 2022

McMaster University, Hamilton, ON

- Investigated resveratrol, caffeine, beta-carotene and EGCG effects on amyloid aggregation in synthetic brain membranes.

Research Assistant, W Booth School of Engineering Practice and Technology | Jan-June 2023

McMaster University, Hamilton, ON

- Built and validated full-stack applications using JavaScript frameworks for academic transcript generation.

Professional & Work Experience

Data and Strategy Intern | June-Aug 2023

McLaren Racing, Arrow McLaren IndyCar, Indianapolis, IN

- Developed modelling pipelines, stochastic simulators and deep-learning computer-vision applications to support race, performance and data engineers.

Lead, Data Analytics and Strategy | Oct 2021-Jan 2024

MAC Formula Electric, Hamilton, ON

- Developed CAN communication protocols, vehicle-control integration and driver-dashboard systems.

Full Stack Engineer | 2020-2021

Syth-Med Biotechnology

- Developed full-stack applications for biotechnology solutions.

DevOps Technical Lead | 2020-2021

WaaW Global Inc.

- Led DevOps infrastructure and technical operations.

Ambassador | 2022-2023

Journal of Science Policy & Governance

- Promoted student engagement in science policy and governance.

Teaching & Mentoring Experience

Undergraduate Teaching Assistant, Physics and Astronomy | Sept 2024-May 2025

McMaster University, Hamilton, ON

- Supported undergraduate physics instruction in introductory mechanics and electricity and magnetism.

Mentor | Sept 2024-May 2025

McMaster Undergraduate Physics Society

- Helped physics students navigate courses, internships and extracurricular opportunities.

Mentor | Sept 2024-May 2025

McMaster Science Careers and Experience Centre

- Supported medical and biological physics co-op students with placement searches, application documents and interview preparation.

Head Yoga Instructor

McMaster University Athletics

- Led yoga instruction for student cohorts; role began in May 2022.

Selected Peer-Reviewed Publications

Ram, U. S., Popple, R. A., Fiveash, J. B., Heinzman, K. A., Kumar, Y., Viscariello, N. N., Stanley, C. B., Stanley, D. N., Cardenas, C. E., & Pogue, J. A. (2025). **Dosimetric evaluation of Ethos 2.0 high-fidelity mode for single-isocenter SRS**. *Journal of Applied Clinical Medical Physics*, 26(12), e70370. doi:10.1002/acm2.70370

Ram, U. S., Pogue, J. A., Soike, M., Pfister, N. T., Jacob, R., & Cardenas, C. E. (2025). **Assessing quantitative performance and expert review of multiple deep learning-based frameworks for computed tomography-based abdominal organ auto-segmentation**. *Intelligent Oncology*, 1(2), 160-171. doi:10.1016/j.intonc.2025.03.003

Gastaldo, I. P., Himbert, S., **Ram, U.**, & Rheinstädter, M. C. (2020). **The effects of resveratrol, caffeine, beta-carotene, and epigallocatechin gallate (EGCG) on amyloid-beta₂₅₋₃₅ aggregation in synthetic brain membranes**. *Molecular Nutrition & Food Research*, 64(22), e2000632. doi:10.1002/mnfr.202000632

Selected Conference Presentations & Talks

Characterizing Contouring Uncertainty across Clinical Specialties in CBCT-Guided Online Adaptive Partial Breast Irradiation | Sept 29, 2026

ASTRO Annual Meeting, Boston, MA · Poster 3104

[ASTRO presenter profile and presentation listing](#)

Improving TG-263 Target Name Compliance Using Locally-Hosted Large Language Models | July 2025

American Association of Physicists in Medicine · Blue Ribbon Poster

Dosimetric Comparison of 6X and 10X Flattening Filter-Free Photon Beams on a TrueBeam Linac Using the Eclipse TPS | July 2025

American Association of Physicists in Medicine · Therapy Poster

Head-to-Head Comparison of State-of-the-Art AutoML Segmentation Frameworks Using Publicly Available Data | July 2024

American Association of Physicists in Medicine · Poster

Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, Single-Isocenter Stereotactic Radiotherapy: A Semi-Automated Comparative Analysis | July 2024

American Association of Physicists in Medicine · Poster

Assessing Efficacy of Multiple, State-of-the-Art Deep Learning Models in Abdominal Organ Auto-Segmentation | July 2023

American Association of Physicists in Medicine · ePoster

Abdominal Multi-Organ Segmentation Using Deep Learning | Aug 2022

Canadian Undergraduate Medical Physics Conference · Talk

Quantitative Analysis of Gel Electrophoresis Images Using Cloud-Based Machine-Learning Analysis | May 2021

Canadian Association of Physicists Conference · Talk

Honors & Awards

- **Science Co-op Student of the Year, Year 5** · McMaster University · 2026
- **Blue Ribbon Poster** · American Association of Physicists in Medicine · 2025
- **Global Experience Award** · McMaster University · 2025
- **Outstanding Poster Presentation** · AAPM / Society of Physics Students · 2024
- **Best Talk, 1st Prize** · Canadian Undergraduate Physics Conference · 2023
- **Academic Entrance Scholarship** · McMaster University, Faculty of Science · 2021
- **Advanced Placement Scholar with Distinction** · College Board · 2021

Selected Appointments & Leadership

UAB-McMaster Ambassador

Mary Heersink Institute for Global Health / McMaster University

- Supported visiting students and trainees, cross-institution communication, outreach and the undergraduate exchange experience.

Additional leadership roles have included President of Hindu YUVA at McMaster, Vice President of Cultural Affairs for Hindu YUVA at UAB, and Vice President, Social, of the McMaster Undergraduate Physics Society.

Community Service & Volunteer Work

Volunteer | Oct 2021-Jan 2024

Humber River Hospital, Toronto, ON

- Supported medical imaging, surgical inpatient and information services by guiding patients and visitors and restocking medical supplies.

Certifications & Training

- Ethos Adaptive Radiotherapy Course · University of Alabama at Birmingham · 2025
- Stereotactic Radiosurgery Course · University of Alabama at Birmingham · 2025
- Advanced Open Water Diver · Professional Association of Diving Instructors · 2025
- Grade C Racing License · Federation of Motor Sports Clubs of India · 2023
- Royal Conservatory of Music · Level 8 Violin and Level 9 Piano · 2019
- Yoga teacher training (RYT-200) · Yoga Alliance · 2019
- French Language Certification (DELF-B2) · 2019

Technical Skills

Programming and software: Python, machine learning, deep learning, computer vision, full-stack engineering, DevOps, MIDAS, Pi Toolbox and SBG RaceWatch.

Research and engineering: Medical physics, radiation oncology, clinical data science, molecular dynamics, genetics, modelling, LLM integration, motorsport analytics, race engineering, performance engineering and augmented reality.

Languages: English (native), Tamil (fluent), French (fluent), Spanish (intermediate).

Open-Source Experience

Beta testing and contributions involving openpilot (comma.ai), Project MONAI and Devin. Public software profile: github.com/udiram.